

All rational cycles of monic integral conservative cubic Hénon maps and the sharp eleven-point bound

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Abstract

We classify every rational periodic orbit of $H_f(x, y) = (y, f(y) - x)$ for every monic cubic $f(t) = t^3 + bt^2 + ct + a$ with $a, b, c \in \mathbb{Z}$. The difficulty is to control the coexistence of different cycles uniformly in all three coefficients, rather than to bound the length of a single cycle. Rational periodic points are integral, and the extrema of the entire periodic coordinate set give an integral third root of a cubic secant remainder. This reduces all sufficiently large diameters to ninety graphs over $\mathbb{Z}[D]$; an exact finite complement contains 9,373 explicitly bounded parameter tuples. We give the full graph procedure and its result tables, with no period or original-coefficient cutoff. Every cycle uses at most three distinct coordinate values and has least period 1, 2, 3, 4, or 6. Seven exact word and polynomial templates describe all cycles. There are at most eleven rational periodic points in total, with equality precisely when $f(t) = (t - h)^3 - 5(t - h) + 2h$ for an integer h . The coordinate set of all coexisting cycles can nevertheless contain more than three values. The theorem uses ordinary iteration and counts points without identifying the two orientations of a cycle.

1 The full cubic classification

The rational periodic points of an individual polynomial automorphism can be finite even when a direct search depends strongly on its coefficients. For the full monic integral conservative cubic family, that dependence can be removed: every cycle has one of seven explicit forms, and the largest possible union of cycles has eleven points. The sharp bound concerns all cycles of the same map, not merely a bound on their individual lengths.

Throughout the paper, fix

$$f(t) = t^3 + bt^2 + ct + a, \quad a, b, c \in \mathbb{Z}, \quad H_f(x, y) = (y, f(y) - x). \quad (1)$$

Its Jacobian determinant is +1 and its inverse is $(x, y) \mapsto (f(x) - y, x)$. The domain is all of $\mathbb{Q}^2$. Let $\mathcal{P}_f = \text{Per}(H_f, \mathbb{Q}^2)$ be the union of its periodic orbits. All periods below are least periods for ordinary iteration by H_f . An orbit with coordinate word $(v_0, \dots, v_{n-1})$ consists of the points (v_i, v_{i+1}) , with indices taken modulo n . The word is identified under cyclic rotation, never under reversal. Its recurrence is

$$v_{i-1} + v_{i+1} = f(v_i). \quad (2)$$

Theorem 1.1 (All coefficients, all rational periodic points). *For every polynomial in (1), the following statements hold.*

1. *Every rational periodic point is integral. Each cycle uses at most three distinct coordinate values, and all its possibilities are the seven rows of Table 1.*
2. *Every least period belongs to $\{1, 2, 3, 4, 6\}$; each of these five periods occurs for a member of the family.*

3. The entire periodic set satisfies $\#\mathcal{P}_f \leq 11$. Equality holds if and only if there is $h \in \mathbb{Z}$ such that

$$f(t) = (t - h)^3 - 5(t - h) + 2h. \quad (3)$$

In the template table, the parameters are integers and every displayed condition is an identity of polynomials, except the fixed-point condition $f(u) = 2u$. For compactness set

$$R_{u,v}(t) = (t - u)(t - v)(t + u + v + b). \quad (4)$$

The b in this expression is the original quadratic coefficient of f . To obtain the cycles of one given polynomial, take the union of all rows whose conditions hold. For the three-distinct-symbol three-cycle, the identity forces $b = -(u + v + w)$; it leaves no extra quadratic coefficient free.

Period	Coordinate word and ranges	Exact condition on $f(t)$
1	$(u), u \in \mathbb{Z}$	$f(u) = 2u$
2	$(u, v), u < v$	$-2t + 2(u + v) + R_{u,v}(t)$
3	$(u, u, v), u \neq v$	$-t + (2u + v) + R_{u,v}(t)$
3	(u, v, w) and $(u, w, v), u < v < w$	$-t + (u + v + w) + (t - u)(t - v)(t - w)$
4	$(u, u, v, v), u < v$	$(u + v) + R_{u,v}(t)$
4	$(h - k, h, h + k, h), h \in \mathbb{Z}, k \geq 1$	$2h + (t - h)((t - h)^2 - k^2)$
6	$(h - k, h - k, h, h + k, h + k, h), h \in \mathbb{Z}, k \geq 1$	$t + h + (t - h)((t - h)^2 - k^2)$

Table 1: The complete template list. In each nonfixed row, the last column equals $f(t)$. The two orientations in the fourth row are different ordinary cycles. The ranges exclude lower-period degeneracies.

The proof has three parts. First, integrality and a real escape bound make the whole periodic set finite. Its two global coordinate extrema then give an integral secant slope and an integral third root of the cubic remainder. Second, the secant inequality forces either an affine recurrence on three roots or one of finitely many endpoint graphs. For large diameters these are exact graphs over $\mathbb{Z}[D]$, not graphs sampled at one large integer. The remaining diameters have proved finite parameter ranges. Third, short-word interpolation gives all templates, and the unique maximizing graph gives (3).

The finite arithmetic tables are part of the proof. Their complete enumeration is specified in Appendix A; every input range is justified before it is used. They involve no floating-point decision, guessed period cutoff, or bounded scan through the original coefficients. The manuscript separates the analytic reduction from the exact finite computation.

2 Arithmetic context and the scope of the increment

General restrictions on integral cycles precede this classification. Pezda studies polynomial mappings of $\mathbb{Z}^2$ and gives, in Theorem 2.1, the cycle-length list

$$\{1, 2, 3, 4, 6, 8, 9, 12, 16, 18, 24\};$$

see Pezda [5]. That is a restriction on cycle length, not the sharp number of periodic points in the particular cubic family (1). We do not claim the existence of a uniform integral period bound

as new. Nor do we use the displayed list as a substitute for the cubic graph argument: the latter proves the smaller period set and the coexistence bound directly.

Canonical heights provide a wider arithmetic setting for polynomial automorphisms. Ingram develops height and specialization results for Hénon maps and explicitly discusses earlier boundedness work of Pezda [3]. Our use of local maxima is a direct elementary argument in a restricted monic integral family, not a new general height theorem. The published bibliographic record is cited; the text used for the comparison is the introduction and opening local-height material of arXiv:1111.3609v1. In particular, the quadratic conjecture for $(y, x + y^2 + b)$ discussed there has determinant -1 . It is not the family studied here. No assertion about that conjecture follows from Theorem 1.1.

The arithmetic questions collected in Section 11 of de Hénon [2] distinguish general rational-coefficient uniform boundedness, interpolation constructions and special determinants. Those questions are broader than the present all-integral cubic classification. Elementary interpolation can produce selected short orbits; it does not on its own exclude additional orbits elsewhere. The main issue here is precisely that exclusion and the resulting coexistence theorem. We therefore deduct word interpolation as a method, while proving its exact seven-template consequence under the global restrictions established below.

Rational coefficients cannot be silently substituted for integral coefficients. The rational-coefficient, integer-valued polynomials of Kim et al. [4] have many periodic points as their odd degrees grow. Their discrete-sine cubic is

$$s_3(t) = \frac{t^3 - 7t}{6},$$

which is neither monic nor in $\mathbb{Z}[t]$. We use the explicitly dated second preprint version for this comparison, not an asserted later journal text. Sharing determinant $+1$ does not make that coefficient family equal to (1).

There is also a nearby repository predecessor: the anonymous C412 manuscript classifies monic integral conservative quadratic Hénon maps and obtains a sharp eight-point bound [1]. It is a repository manuscript, not a journal publication. Its monic integrality argument, coordinate-word encoding and finite-complement strategy are not new contributions of this paper. In the cubic problem, the third secant root creates additional branches and allows different three-symbol cycles to coexist. The residual result is the complete closure of that cubic coexistence problem, including every original coefficient triple and the entire eleven-point equality locus.

Comparison	Established or accessed statement	Distinction here
Pezda	General integral cycle-length restrictions	Sharp total points in one fixed cubic family, with exact templates
Ingram and the question survey	Heights, broader arithmetic questions and interpolation context	All integral coefficients, degree three, determinant $+1$, and a direct exhaustive proof
Kim–Krieger–Postolache–Szeto	Rational-coefficient integer-valued examples of growing degree	Leading coefficient exactly $+1$ and all coefficients integral
C412 repository manuscript	Complete conservative quadratic classification; sharp eight points	Integral third secant root, cubic coexistence, sharp eleven points

Table 2: Scope comparison, not a claim of exhaustive literature priority. The present proof does not invoke an uninspected external theorem to perform its finite reduction.

These comparisons are bounded by the primary texts actually accessed. No global-priority guarantee is asserted. The theorem makes no claim for nonmonic cubics, arbitrary rational

coefficients, determinant -1 , number fields, positive characteristic or arbitrary degree.

3 Integrality and the global secant

The normalization in this section uses the coordinate set of all periodic points of one map. Choosing the extrema separately for individual cycles would not control their total number.

Lemma 3.1 (Integral points and a common finite box). *Every point in $\mathcal{P}_f$ is integral, and*

$$\mathcal{P}_f \subseteq ([-R, R] \cap \mathbb{Z})^2, \quad R = |a| + |b| + |c| + 2.$$

In particular, $\mathcal{P}_f$ is finite.

Proof. Take any periodic coordinate sequence (x_i) satisfying (2). For a prime p , suppose $T = \max_i |x_i|_p > 1$. At an index attaining T , the leading term x_i^3 has p -adic absolute value T^3 . The lower terms have absolute values at most T^2 , T and 1 , since the coefficients are integral. The ultrametric inequality with a unique largest term therefore gives $|f(x_i)|_p = T^3$. But $|x_{i-1} + x_{i+1}|_p \leq T$, a contradiction. This holds at every prime, so all rational coordinates are integers.

At the real place let $M = \max_i |x_i|$. From the recurrence at an index attaining M ,

$$M^3 \leq |b|M^2 + (|c| + 2)M + |a|.$$

If $M \geq 1$, the right side is at most RM^2 , so $M \leq R$. If $M < 1$, the same bound holds since $R \geq 2$. The bound is common to every orbit of this polynomial. Thus the entire periodic set lies in the stated finite lattice square. $\square$

If $\mathcal{P}_f$ is empty, the bounds in Theorem 1.1 hold without further work. If the union E of all its coordinate values has one element u , the periodic set is just the diagonal fixed point (u, u) . Assume from now on that E has at least two elements, and set

$$m = \min E, \quad D = \max E - m \geq 1, \quad g(t) = f(t + m) - 2m. \quad (5)$$

Writing $T_m(x, y) = (x + m, y + m)$, we have $H_g = T_{-m}H_fT_m$. The translated global coordinate set $E' = E - m$ lies in $\{0, \dots, D\}$ and contains both endpoints. The polynomial g remains monic and integral. This is an internal conjugacy, not a restriction on the original quadratic coefficient.

Lemma 3.2 (Integral secant root). *There are integers A, q, r such that*

$$g(t) = A + qt + t(t - D)(t - r), \quad (6)$$

$$-2 \leq q \leq 2, \quad 0 \leq A \leq 2D, \quad 0 \leq A + qD \leq 2D. \quad (7)$$

Every $t \in E'$ satisfies

$$|t(t - D)(t - r)| \leq 2D. \quad (8)$$

Proof. When t occurs in a cycle, $g(t)$ is the sum of its two neighbors, both in $[0, D]$. Hence $0 \leq g(t) \leq 2D$ for every $t \in E'$. Set $A = g(0)$ and $q = (g(D) - g(0))/D$. Since g has integer coefficients, D divides $g(D) - g(0)$; the endpoint bounds give (7). The monic cubic $g(t) - A - qt$ vanishes at 0 and D . If its quadratic coefficient is B , its third root is $r = -B - D \in \mathbb{Z}$, proving (6).

The line $A + qt$ interpolates the endpoint values $g(0)$ and $g(D)$. It belongs to $[0, 2D]$ throughout $[0, D]$. For an actual coordinate $t \in E'$, both $g(t)$ and this line value are in that interval. Their difference is the cubic product in (6), giving (8). $\square$

Lemma 3.3 (The affine-root case). *If $E' \subseteq \{0, D, r\}$, there are at most nine periodic points. Every least period is in $\{1, 2, 3, 4, 6\}$, and each cycle uses at most three symbols. If $E' \subseteq \{0, D\}$, the point bound is four, irrespective of r .*

Proof. At a root of the cubic remainder the recurrence becomes

$$x_{i+1} = A + qx_i - x_{i-1}. \quad (9)$$

At most three coordinate values give at most nine ordered pairs. For $q \neq 2$, subtract the stationary value $A/(2 - q)$; the resulting homogeneous pair recurrence has matrix

$$L_q = \begin{pmatrix} 0 & 1 \\ -1 & q \end{pmatrix}.$$

Multiplication gives $L_{-1}^3 = I$, $L_0^4 = I$ and $L_1^6 = I$. The least periods divide the indicated orders. For $q = -2$ the homogeneous coordinate solutions are $(\alpha + \beta i)(-1)^i$; a periodic solution must have $\beta = 0$, so its period divides two. For $q = 2$, summing (9) around a cycle gives $nA = 0$ and hence $A = 0$. Consecutive differences are then constant; summing them over one period makes that constant zero. The sequence is fixed. The union of all these possible divisors is $\{1, 2, 3, 4, 6\}$. With only two symbols there are at most four ordered pairs, and the same recurrence applies. $\square$

The last assertion includes $D = 1$. The one-symbol and empty cases already cover $D = 0$ and absence of extrema. These cases are not sent to a computation that assumes an interior symbol.

4 Uniform graphs for large diameter

Lemma 4.1 (The five endpoint alphabets). *Suppose $D \geq 10$ and $E' \not\subseteq \{0, D, r\}$. After the reflection $t \mapsto D - t$ if necessary, $r \in \{-1, 0, 1, 2, 3\}$ and $E' \subseteq S_r = T_r \cup \{D\}$, where the sets T_r are listed in Table 3.*

Proof. For an extra symbol $t \notin \{0, D, r\}$, integrality gives $|t - r| \geq 1$. If $3 \leq t \leq D - 3$, then

$$|t(t - D)(t - r)| \geq t(D - t) \geq 3(D - 3) > 2D,$$

contrary to (8). Thus the only possible extra symbols are $1, 2, D - 2, D - 1$. A lower extra at 1 or 2 satisfies respectively

$$|1 - r| \leq \left\lfloor \frac{2D}{D - 1} \right\rfloor = 2, \quad |2 - r| \leq \left\lfloor \frac{D}{D - 2} \right\rfloor = 1.$$

Their union forces $r \in \{-1, 0, 1, 2, 3\}$. Upper extras force $D - r$ to lie in the same set. These two ranges are disjoint for $D \geq 10$.

Reflection in both coordinates conjugates H_g to $H_{\tilde{g}}$, where a direct expansion yields

$$\begin{aligned} \tilde{g}(z) &= 2D - g(D - z) = \tilde{A} + qz + z(z - D)(z - \tilde{r}), \\ \tilde{r} &= D - r, \quad \tilde{A} = (2 - q)D - A. \end{aligned}$$

It preserves the monic integral family, the ordinary clock and all point counts. We may therefore use the lower range. Retaining its roots inside $[0, D]$ and its allowed extras gives $T_{-1} = T_0 = \{0, 1\}$, $T_1 = T_2 = \{0, 1, 2\}$ and $T_3 = \{0, 1, 2, 3\}$. If neither endpoint range occurs, there were no extra symbols and Lemma 3.3 already applies. $\square$

Write $T_r = \{0, \dots, m_r\}$, so m_r is 1, 2 or 3. For $D \geq 16$, every sum of two symbols in S_r has a unique representation $kD + s$. Its coefficient pair belongs to

$$\Sigma_r = \{(0, s) : 0 \leq s \leq 2m_r\} \cup \{(1, s) : 0 \leq s \leq m_r\} \cup \{(2, 0)\}. \quad (10)$$

Since 0, D both occur in the actual coordinate set, the endpoint images are sums of two symbols. Thus

$$A = kD + s, \quad (k, s) \in \Sigma_r, \quad (k + q, s) \in \Sigma_r. \quad (11)$$

To justify the last expression, representations with constant terms between 0 and $2m_r \leq 6$ cannot differ by a nonzero multiple of D . The same no-cancellation argument will justify every graph edge.

Lemma 4.2 (No cancellation between the two coefficients). *For each case in (11) and every $D \geq 16$, the partial graph of H_g on S_r^2 is independent of D when symbols are represented by their coefficient pairs in $\mathbb{Z}[D]$.*

Proof. For $t \in T_r$, expansion of (6) gives

$$g(t) = [k - t(t - r)]D + [s + qt + t^2(t - r)], \quad (12)$$

whereas $g(D) = (k + q)D + s$. An edge $(x, t) \mapsto (t, z)$ is present precisely when $g(t) = x + z$. The constant coefficient of $x + z$ belongs to $[0, 2m_r]$. Allowing $s, u \in [0, 2m_r]$, $q \in [-2, 2]$ and $t \in T_r$, the absolute constant discrepancy

$$|s + qt + t^2(t - r) - u|$$

is at most 6, 5, 12, 8, 14 for $r = -1, 0, 1, 2, 3$, respectively. These bounds follow by substituting the two, three or four listed values of t and the endpoints of the ranges for s, u, q . For $t = D$ the bound is at most six. Every bound is less than D . If the coefficient of D in an edge discrepancy were a nonzero integer, its magnitude at this D would be at least D , which the constant discrepancy cannot cancel. Hence numeric equality is equivalent to equality of both coefficients, as claimed. $\square$

This lemma concerns literal affine polynomials, not empirical stability at one chosen diameter. It leaves only ninety finite graphs. Indeed, for $s = 0$ there are nine pairs of allowed values of k and $k + q$; for each $1 \leq s \leq m_r$ there are four; and for each $m_r < s \leq 2m_r$ there is one. The count is $9 + 5m_r$, giving the Cases column of Table 3.

Proposition 4.3 (The complete large graph computation). *The graphs in (11) have the data in Table 3. Every cycle uses at most three symbols and has least period in $\{1, 2, 3, 4\}$. Each graph has at most eight periodic vertices. Including the affine-root cases, every map whose global diameter is at least sixteen has at most nine periodic points.*

Proof. Use the coefficient pairs $(0, t)$ for $t \in T_r$ and $(1, 0)$ for D . Enumerate (11), evaluate (12), and extract all cycles of $(x, y) \mapsto (y, g(y) - x)$ by coefficientwise arithmetic. Appendix A specifies every loop and the complete cycle-extraction algorithm, and proves its correctness. Its exact evaluation gives Table 3.

For a compact explicit enumeration of every endpoint-using graph, Table 4 lists all its cycles. There are five baseline cases for each of the five r values, together with four additional cases, giving $25 + 4 = 29$ endpoint-using cases. Direct substitution into (6) verifies their displayed words; the full graph procedure also excludes any omitted cycles. Cases without both ends are included in the first table and its algorithm. Lemma 4.1 covers every non-affine actual map, and Lemma 4.2 identifies its graph uniformly in D . Lemma 3.3 supplies the remaining nine-point bound. $\square$

r	T_r	Cases	Both ends	Max. points	Periods	Max. symbols
-1	$\{0, 1\}$	14	5	4	1, 2, 3, 4	2
0	$\{0, 1\}$	14	5	4	1, 2, 3, 4	2
1	$\{0, 1, 2\}$	19	6	6	1, 2, 3, 4	3
2	$\{0, 1, 2\}$	19	7	8	1, 2, 3, 4	3
3	$\{0, 1, 2, 3\}$	24	6	6	1, 2, 3, 4	3

Table 3: All ninety symbolic graphs. “Both ends” counts cases in which the union of graph cycles uses 0 and D . The maxima and period lists include all cases, even those not using both ends. “Max. symbols” is per cycle, not for the union.

r	A	q	All graph cycles
All five	0	2	$(0), (D)$; also (r) for $r = 1, 2, 3$
All five	D	-1	$(0, 0, D)$
All five	D	0	$(0, 0, D, D)$
All five	$2D$	-2	$(0, D)$
All five	$2D$	-1	$(0, D, D)$
1	$D + 1$	-1	$(0, 1, D), (0, D, 1)$
2	2	0	$(1, 1, D), (0, 0, 2, 2)$
2	$D + 2$	-1	$(1, D), (0, 2, D), (0, D, 2)$
3	$D + 3$	-1	$(0, 3, D), (0, D, 3)$

Table 4: Every symbolic case whose graph cycles use both endpoints. “All five” means each $r \in \{-1, 0, 1, 2, 3\}$. These are cycles inside the indicated alphabet; the table does not by itself exclude periodic coordinates outside $[0, D]$ for a polynomial chosen from one of its rows.

Remark 4.4 (Four global symbols coexist). For any integer $D \geq 16$, the polynomial

$$g_D(t) = 2 + t(t - D)(t - 2)$$

has the disjoint cycles $(1, 1, D)$ and $(0, 0, 2, 2)$, of least periods three and four. Indeed $g_D(0) = g_D(2) = 2$ and $g_D(1) = D + 1$, $g_D(D) = 2$. Their seven points already use the four distinct values $0, 1, 2, D$. Thus the full coordinate set need not have at most three elements. No assertion about the absence of additional cycles is needed for this counterexample.

The endpoint condition in the graph tables is only a necessary filter for a global-extrema tuple. It does not certify that a polynomial has no cycles outside the displayed interval. For example, the baseline $r = -1$, $A = 0$, $q = 2$ also has the fixed coordinate -1 , which lies outside its chosen graph. When the tables are applied to our original map, this issue cannot hide a cycle: $0, D$ were chosen from the entire finite periodic set in (5).

5 The exact finite complement

Only diameters $2 \leq D \leq 15$ remain. If there is no interior coordinate, Lemma 3.3 already gives the two-symbol bound. Otherwise some $t \in E' \cap \{1, \dots, D - 1\}$ exists, and (8) implies

$$|t - r| \leq \frac{2D}{t(D - t)} \leq \frac{2D}{D - 1} \leq 4.$$

Consequently $-3 \leq r \leq D + 3$. Combining this with the endpoint conditions gives the complete finite ranges

$$\begin{aligned} 2 \leq D \leq 15, \quad -3 \leq r \leq D + 3, \quad 0 \leq A \leq 2D, \\ q \in \{-2, -1, 0, 1, 2\}, \quad 0 \leq A + qD \leq 2D. \end{aligned} \tag{13}$$

This bounds the normalized secant parameters after the global reduction; it does not impose a cutoff on a, b, c .

For each tuple define

$$S = \{t \in \mathbb{Z} : 0 \leq t \leq D, |t(t - D)(t - r)| \leq 2D\} \tag{14}$$

and the partial map on S^2

$$F(x, y) = (y, A + qy + y(y - D)(y - r) - x). \tag{15}$$

An image leaving S^2 has no outgoing graph edge. Every actual periodic point of a global-extrema tuple lies on a cycle of this graph. Conversely, every graph cycle is an actual polynomial orbit. Retain a tuple only if the union of its graph cycles uses both 0 and D . This is a necessary condition for an actual global-extrema tuple, and hence cannot discard the periodic set of any map under consideration.

Proposition 5.1 (Finite arithmetic certificate). *The complete computation for (13) gives Table 5. Every retained graph cycle uses at most three symbols. Its least period belongs to $\{1, 2, 3, 4, 6\}$. The largest total number of graph-periodic vertices is eleven, attained at the single tuple*

$$(D, r, A, q) = (4, 2, 6, -1). \tag{16}$$

The cycles of this tuple are

$$(2), \quad (0, 3), \quad (1, 4), \quad (0, 2, 4), \quad (0, 4, 2). \tag{17}$$

D	Compatible tuples	Both graph ends	Maximum points	Periods
2	117	51	9	1, 2, 3, 4, 6
3	170	56	9	1, 2, 3, 4
4	231	63	11	1, 2, 3, 4, 6
5	300	68	8	1, 2, 3, 4
6	377	75	9	1, 2, 3, 4, 6
7	462	78	8	1, 2, 3, 4
8	555	85	9	1, 2, 3, 4, 6
9	656	90	8	1, 2, 3, 4
10	765	97	9	1, 2, 3, 4, 6
11	882	102	8	1, 2, 3, 4
12	1007	109	9	1, 2, 3, 4, 6
13	1140	114	8	1, 2, 3, 4
14	1281	121	9	1, 2, 3, 4, 6
15	1430	126	8	1, 2, 3, 4

Table 5: The full small-diameter complement: 9,373 tuples satisfy the endpoint value inequalities and 1,235 use both endpoints among graph cycles. Point maxima and period lists are taken after this necessary filter. In every row the maximum number of symbols per cycle is three.

Proof. For each tuple in (13), form (14), evaluate (15) exactly, and extract every directed cycle. Retain precisely those tuples passing the endpoint filter. Sum the lengths of the distinct cycles to count periodic vertices. Appendix A supplies the complete integer pseudocode, including the extraction algorithm, the filter, the maximizer update and the period and symbol statistics. Its exact evaluation is Table 5, (16) and (17).

The first count in each row can also be checked directly. There are $D + 7$ choices for r . At $q = -2, -1, 0, 1, 2$, the numbers of allowed A values are respectively 1, $D + 1$, $2D + 1$, $D + 1$, 1. Their sum is $4D + 5$, giving $(D + 7)(4D + 5)$ tuples. The graph algorithm, rather than this input count, establishes every cycle statistic.

All arithmetic decisions are integer equalities or inequalities. The graph is finite, and the extraction lemma in the appendix proves that it records every cycle exactly once and records its least period. No assumed upper bound eleven enters the computation of the maximum; the maximum and all tuples attaining it are accumulated from the graphs. The reduction preceding (13) proves that this finite computation covers the entire remaining family. $\square$

Together, Lemmas 3.1–3.3 and Propositions 4.3–5.1 prove the eleven-point bound, the possible period list and the per-cycle three-symbol restriction. Their roles are different: the analytic arguments prove the finite reduction, while the exact certificate evaluates its explicitly bounded graphs. The next section uses these established restrictions to derive the full coefficient conditions without any bound on the original coefficients.

6 Exhaustion of the seven templates

A least-period- n orbit has n distinct ordered coordinate pairs. This elementary observation restricts the possible words before interpolation is applied.

6.1 One and two coordinate values

A one-symbol word is fixed, and its condition is $f(u) = 2u$. With two symbols there are only four ordered pairs, so there is no least-period-six cycle. Period two has the word (u, v) , where $u \neq v$; choose $u < v$ to avoid duplicate rotations. The equations $f(u) = 2v$ and $f(v) = 2u$ say that the monic cubic $f(t) + 2t - 2(u + v)$ vanishes at u and v . Dividing by $(t - u)(t - v)$ leaves a monic linear factor. Comparing its quadratic coefficient with b forces that factor to be $t + u + v + b$. This proves the second row of Table 1.

A two-symbol three-cycle has a uniquely repeated symbol, so its word can be written (u, u, v) with $u \neq v$. The recurrence gives $f(u) = u + v$ and $f(v) = 2u$. Consequently $f(t) + t - (2u + v)$ vanishes at both u, v and has the same forced third factor, proving the third row.

A two-symbol four-cycle must have two occurrences of each letter. Indeed a word (u, u, u, v) would force both $f(u) = 2u$ and $f(u) = u + v$, and the analogous assertion holds when the other symbol is repeated three times. With two occurrences of each, the alternating word has period two. The remaining word is (u, u, v, v) up to rotation, and reversal is already a rotation of this word. Its equations are $f(u) = f(v) = u + v$. Factoring $f(t) - (u + v)$ proves the fifth row.

6.2 Three coordinate values and an integer interpolation lemma

If a three-cycle uses distinct symbols $u < v < w$, the neighbors of each are the other two. With $S = u + v + w$, the monic polynomial $f(t) + t - S$ vanishes at all three symbols. It is therefore exactly $(t - u)(t - v)(t - w)$. Both words (u, v, w) and (u, w, v) satisfy the recurrence, and no rotation identifies them. This proves the fourth row, including $b = -S$.

Lemma 6.1 (Integral reciprocal sum). *If p, q are distinct nonzero integers and $1/p + 1/q$ is an integer, then $p + q = 0$.*

Proof. For opposite signs the absolute value of the sum is strictly less than one. If it is an integer it must be zero. For the same sign, change both signs if necessary and arrange $0 < p < q$. When $p = 1$, the sum is strictly between one and two. When $p \geq 2$, it is strictly between zero and one. Neither alternative is integral. $\square$

Consider a three-symbol four-cycle. The repeated symbol cannot occur adjacently: in (h, h, u, v) , the two equations for h would give $h + v = h + u$, contrary to $u \neq v$. The word is therefore (h, u, h, v) , with three distinct integers, and its conditions are

$$f(h) = u + v, \quad f(u) = f(v) = 2h.$$

Let $Q(t)$ be the quadratic interpolant at h, u, v . Since the difference between the monic cubic and this interpolant is

$$f(t) - Q(t) = (t - h)(t - u)(t - v),$$

the quadratic coefficient of Q is an integer. Subtracting the constant $2h$ in its interpolation data shows that this coefficient is

$$\frac{u + v - 2h}{(h - u)(h - v)} = \frac{1}{u - h} + \frac{1}{v - h}.$$

Lemma 6.1 gives $u + v = 2h$. Thus the two endpoints are $h - k, h + k$ with $k \geq 1$, and $Q(t) = 2h$. The resulting monic identity is exactly the centered four-cycle row in the table. Swapping the two endpoints only rotates the displayed word.

6.3 The three-symbol six-cycle

First suppose that a three-symbol six-cycle has no equal adjacent letters. Its six distinct ordered pairs must be all six off-diagonal pairs on the alphabet. Hence each symbol occurs twice, with the other two symbols each occurring once as predecessor and once as successor. At a symbol u , equality of the two neighbor sums has two possible pairings of predecessors and successors. Pairing each predecessor with itself would require $2v = 2w$ for the other distinct symbols, which is impossible over $\mathbb{Z}$. Both occurrences must therefore have the unordered neighbor pair $\{v, w\}$. At every step, after two distinct symbols the next symbol is the third one. The word repeats the same order of three symbols and has period three, a contradiction.

There must be adjacent equal letters p, p . The equations at the two occurrences force the predecessor and successor of this pair to be the same symbol, say h . They cannot be p : a triple p, p, p gives $f(p) = 2p$ and forces its neighbors successively to equal p , making the entire cycle constant. Rotate the word to $(h, p, p, h, *, *)$ with $h \neq p$. The third symbol q must occur in one or both of the remaining positions. The possibilities involving q are

$$(p, q), (q, p), (h, q), (q, h), (q, q).$$

In the first two, comparing the additional occurrence of p with an occurrence in the double forces $q = p$. In the next two, the equations at the additional h force $q = h$. These contradict the distinctness of the third symbol. The last two positions are therefore (q, q) , giving the word

$$(p, p, h, q, q, h).$$

Its recurrence equations are

$$f(p) = p + h, \quad f(q) = q + h, \quad f(h) = p + q.$$

Apply interpolation to the monic cubic $f(t) - (t + h)$. It has values zero at p, q and value $p + q - 2h$ at h . As in the four-cycle argument, the quadratic coefficient of its quadratic interpolant is an integer and equals

$$\frac{p + q - 2h}{(h - p)(h - q)} = \frac{1}{p - h} + \frac{1}{q - h}.$$

Lemma 6.1 forces $p + q = 2h$. Writing the endpoints as $h - k, h + k$ now gives

$$f(t) = t + h + (t - h)((t - h)^2 - k^2),$$

and the displayed period-six word in Table 1. Swapping endpoints rotates this word by three places.

6.4 Converses and exact least periods

The previously proved period and symbol restrictions exhaust all cases considered above. Conversely, substituting each of the seven conditions into (2) verifies its word. The ranges make the ordered pairs distinct. The two-cycle and the three-distinct-symbol three-cycle have distinct successive first coordinates. The word (u, u, v) uses $(u, u), (u, v), (v, u)$, which are distinct for $u \neq v$. The two-symbol four-cycle uses $(u, u), (u, v), (v, v), (v, u)$; and the centered four-cycle uses $(h - k, h), (h, h + k), (h + k, h), (h, h - k)$. For the six-cycle, subtracting h from both coordinates gives the pairs

$$(-k, -k), (-k, 0), (0, k), (k, k), (k, 0), (0, -k),$$

which are distinct for $k \geq 1$. Their asserted lengths are therefore their least periods. Rows with the same period but different numbers of symbols cannot describe the same orbit.

All five least periods occur. The polynomial t^3 has a fixed point and the centered six-cycle with $h = 0, k = 1$. The polynomial $t^3 - 3t$ has the two-cycle $(-1, 1)$; $t^3 - 2t$ has the three-cycle $(-1, 0, 1)$; and $t^3 - t$ has the four-cycle $(-1, 0, 1, 0)$. This completes the template and period parts of Theorem 1.1.

7 The entire equality locus and ordinary returns

Proof of the equality assertion in Theorem 1.1. The affine-root and large-diameter cases have at most nine points. The cases using only two symbols have at most four. Thus a map with eleven periodic points must have its global normalization at the unique tuple (16). The normalized polynomial is

$$g(t) = 6 - t + t(t - 4)(t - 2) = t^3 - 6t^2 + 7t + 6 = (t - 2)^3 - 5(t - 2) + 4.$$

Undoing $g(t) = f(t + m) - 2m$ and putting $h = m + 2$ gives precisely (3). In the original coefficients it is

$$b = -3h, \quad c = 3h^2 - 5, \quad a = -h^3 + 7h. \quad (18)$$

This proves necessity for every original coefficient triple.

Conversely, for $f_0(t) = t^3 - 5t$ the disjoint cycles are

$$(0), \quad (-2, 1), \quad (-1, 2), \quad (-2, 0, 2), \quad (-2, 2, 0). \quad (19)$$

Substitution verifies them, and their lengths 1, 2, 2, 3, 3 give eleven distinct points. The proved upper bound excludes any additional rational periodic points. Conjugating by T_h translates every coordinate in (19) by h and gives exactly (3). Hence every integer h realizes equality, with no omitted exceptional coefficient branch. $\square$

The ordinary return series follows from the cycle classification; it is not a separate arithmetic counting problem. Let $c_j(f)$ be the number of distinct cycles of least period j , where $j \in \{1, 2, 3, 4, 6\}$, as obtained from Table 1. Cycles related only by reversal remain distinct when reversal is not already a rotation. Define

$$N_n(f) = \# \text{Fix}(H_f^n, \mathbb{Q}^2), \quad Z_f(z) = \exp \left(\sum_{n \geq 1} \frac{N_n(f)}{n} z^n \right).$$

These counts use actual rational points, each with weight one. Every point fixed by an iterate is periodic, so all counts are finite.

Corollary 7.1 (Native ordinary zeta). *As identities of formal power series,*

$$N_n(f) = \sum_{\substack{j \in \{1, 2, 3, 4, 6\} \\ j|n}} j c_j(f), \quad Z_f(z) = \prod_{j \in \{1, 2, 3, 4, 6\}} (1 - z^j)^{-c_j(f)}.$$

On the eleven-point equality family,

$$Z_f(z) = \frac{1}{(1 - z)(1 - z^2)^2(1 - z^3)^2}.$$

Proof. A cycle of length j contributes all of its j points to the fixed set of the n th iterate exactly when $j \mid n$. This proves the first identity. Its contribution to the logarithm of the zeta is

$$\sum_{m \geq 1} \frac{j}{jm} z^{jm} = \sum_{m \geq 1} \frac{z^{jm}}{m} = -\log(1 - z^j).$$

There are finitely many cycles, so summing and exponentiating gives the product. Formula (19) gives the last identity. $\square$

No scheme multiplicity or auxiliary return clock enters this corollary. The map is invertible, so a rational point that is eventually periodic is already periodic: applying the inverse of an initial iterate removes any purported preperiod.

8 Reproducibility and limitations

Theorem 1.1 closes the full monic integral conservative cubic question in ordinary rational dynamics. The integer third secant root controls every coexisting cycle in one common normalized graph. The finite complement then singles out one normalized eleven-point graph, and short-word interpolation translates the graph restrictions into exact coefficient identities.

The finite certificate is an explicit integer computation after a proved reduction, not a proof-assistant formalization. The retained producer evaluates the ninety symbolic graphs and the 9,373 bounded small-diameter tuples using path traversal. Before manuscript drafting, a separate internal implementation reconstructed the small computation on the full square $\{0, \dots, D\}^2$, without the secant-symbol filter, and isolated the periodic vertices by iterating a partial injection with repeated squaring. Its large-graph arithmetic represented affine polynomials by their values at zero and one; equality of these two values is exact because the expressions were already proved affine. That independent reconstruction agreed with all table entries and the unique maximizing tuple. The present typeset specification is complete without either implementation; Appendix A states the calculation and its correctness argument. This internal verification is not described as human peer review.

Neither the threshold sixteen nor the number of finite cases is claimed to be minimal. Those choices give a short no-cancellation proof and an explicitly bounded complement. A purely hand-classified replacement for the small graph table could change the form of the proof, but is not assumed here. The source comparisons remain bounded by the accessed primary records and the named repository predecessor.

The restrictions on coefficients and determinant are essential to the stated theorem. No extension to all rational-coefficient cubics, other degrees, number fields or positive characteristic is proved. Finally, the finite ordinary zeta is a source-dynamical invariant. It does not establish rational-prime primitive ownership, target Euler factors, root numbers, automorphy, a target zero or divisor correspondence, or a Hilbert–Pólya operator.

A Complete finite-graph specification

This appendix gives the full finite arithmetic specification for Propositions 4.3 and 5.1. The pseudocode uses exact integers, finite sets, dictionaries and lexicographic order. Its half-open `range(a,b)` contains $a, a + 1, \dots, b - 1$; `product` is Cartesian product. There is no imposed orbit-length bound. A word is canonicalized by rotation only.

A.1 Cycle extraction and its correctness

For an ordered finite symbol set S and values $V[y]$, the graph has the edge $(x, y) \mapsto (y, V[y] - x)$ when the last coordinate belongs to S . The symbols and values are either integers or pairs of integer coefficients. Subtraction is ordinary subtraction in the first case and componentwise subtraction in the second.

```
from itertools import product

def minus(u, v):
    if isinstance(u, tuple):
        return (u[0] - v[0], u[1] - v[1])
    return u - v

def cycles(S, V):
    S = tuple(sorted(S))
    allowed = set(S)
    seen = set()
    answer = []
    for start in product(S, repeat=2):
        if start in seen:
            continue
        point = start
        path = []
        position = {}
        while point not in seen:
            if point in position:
                orbit = path[position[point]:]
                word = tuple(x for (x, y) in orbit)
                word = min(word[i:] + word[:i]
                           for i in range(len(word)))
                answer.append(word)
                break
            position[point] = len(path)
            path.append(point)
            x, y = point
            z = minus(V[y], x)
            if z not in allowed:
                break
            point = (y, z)
        seen.update(path)
    return sorted(answer, key=lambda w: (len(w), w))
```

Lemma A.1 (Exact extraction). *The procedure `cycles` terminates and returns every directed cycle of the partial graph once, and no other words. Each returned word has length equal to its least ordinary period.*

Proof. The graph has finitely many vertices. Within one traversal, either an image leaves the graph, the path reaches an already completed traversal, or a vertex repeats on the current path; one of

these events occurs after finitely many steps. A repeat on the current path closes exactly its segment from the first occurrence to the repeat. Its vertices are distinct before closure, so it is one whole cycle with its least period.

Consider any cycle and the first traversal that reaches it. No vertex of that cycle is yet in a completed traversal. Following the deterministic edges therefore closes and records it. A subsequent path reaching that cycle stops at a completed vertex and cannot record it again. Distinct cycles cannot intersect in a functional graph. Marking every traversed vertex completed thus loses no unrecorded cycle and creates no duplicate. The outer loop visits every vertex, so every cycle is eventually reached. Canonical rotation changes only its chosen starting coordinate, not its orientation or its length. $\square$

A.2 Small-diameter enumeration

The following code computes the input counts, the endpoint-filtered point maxima, period sets, per-cycle symbol maxima, and every global maximizer. The value eleven is not supplied as an input or assertion.

```
def small_certificate():
    rows = []
    global_maximum = -1
    equality = []
    all_periods = set()
    for D in range(2, 16):
        cases = kept = 0
        maximum = 0
        max_symbols = 0
        periods = set()
        for r, A, q in product(range(-3, D + 4),
                               range(2 * D + 1),
                               range(-2, 3)):
            if not (0 <= A + q * D <= 2 * D):
                continue
            cases += 1
            S = [t for t in range(D + 1)
                  if abs(t * (t-D) * (t-r)) <= 2 * D]
            V = {t: A + q*t + t*(t-D)*(t-r) for t in S}
            C = cycles(S, V)
            used = {t for w in C for t in w}
            if not (0 in used and D in used):
                continue
            kept += 1
            count = sum(len(w) for w in C)
            maximum = max(maximum, count)
            for w in C:
                periods.add(len(w))
                all_periods.add(len(w))
                max_symbols = max(max_symbols, len(set(w)))
            if count > global_maximum:
                global_maximum = count
                equality = []
            if count == global_maximum:
                equality.append((D, r, A, q, C))
        rows.append((D, cases, kept, maximum,
                    sorted(periods), max_symbols))
    return rows, global_maximum, equality, sorted(all_periods)
```

The ranges are exactly (13); the symbol filter is (14). A retained case has at least one cycle, so every per-cycle maximum is defined. The output rows are Table 5; the global maximizer and its words are (16)–(17). The period union is $\{1, 2, 3, 4, 6\}$ and every row has symbol maximum three.

A.3 Symbolic large-diameter enumeration

Represent $kD + s$ by (k, s) . The symbol D is $(1, 0)$ and an integer symbol $t \in T_r$ is $(0, t)$. Formula (12) defines its image coefficients. The extra statistic **error** measures the maximum absolute constant discrepancy among all edge tests. This statistic checks the arithmetic bound, but the independent proof of uniformity remains Lemma 4.2.

```
def symbolic_certificate():
    rows = []
    endpoint_patterns = []
    for r in (-1, 0, 1, 2, 3):
        T = [t for t in range(4)
              if t == 0 or abs(t * (t-r)) <= 2]
        S = [(0, t) for t in T] + [(1, 0)]
        Sigma = {(u[0] + v[0], u[1] + v[1])
                  for u, v in product(S, repeat=2)}
        cases = kept = maximum = max_symbols = error = 0
        periods = set()
        for (k, s), q in product(sorted(Sigma), range(-2, 3)):
            if (k + q, s) not in Sigma:
                continue
            cases += 1
            V = {(0, t): (k - t*(t-r),
                          s + q*t + t*t*(t-r)) for t in T}
            V[(1, 0)] = (k + q, s)
            for y, u, v in product(S, repeat=3):
                error = max(error, abs(V[y][1]-u[1]-v[1]))
            C = cycles(S, V)
            count = sum(len(w) for w in C)
            maximum = max(maximum, count)
            for w in C:
                periods.add(len(w))
                max_symbols = max(max_symbols, len(set(w)))
            used = {t for w in C for t in w}
            if (0, 0) in used and (1, 0) in used:
                kept += 1
            endpoint_patterns.append((r, (k, s), q, C))
        rows.append((r, T, cases, kept, maximum,
                    sorted(periods), max_symbols, error))
    return rows, endpoint_patterns
```

The five values of **error** are 4, 3, 8, 8, 14, all less than sixteen. They are no larger than the deliberately looser analytic bounds 6, 5, 12, 8, 14. The other row entries give Table 3. Decoding $(0, t)$ as t and $(1, 0)$ as D , and changing only cyclic starting positions, gives exactly the 29 cases summarized in Table 4.

Both enumeration procedures call the same completely specified cycle routine. Every vertex, parameter tuple and comparison is finite and explicit. Sections 3–5 prove why no original polynomial escapes their ranges or the separate affine cases. This is the required distinction between an exhaustive finite certificate and a coefficient experiment.

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